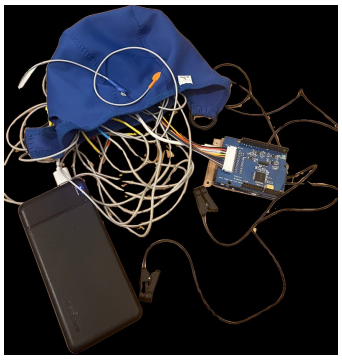
QUICK START & SAFETY GUIDE · SKU CQA240229AR

From the Box to Live EEG over Wi-Fi

Seven steps: find your computer's IP address, upload the sketch with your Wi-Fi details, mount the shield, connect the electrodes, power from a battery, run the Python receiver and check the signal.



WARNING Battery only.

ardEEG has **no isolation circuitry**. Connect the Arduino to a computer only to upload the sketch, with no electrodes on anyone. While recording it runs from a **small 5 V power bank**. ardEEG is **not a medical device**.

01 SAFETY RULES — READ BEFORE PUTTING ELECTRODES ON ANYONE

#	Rule
1	**Battery only.** While electrodes are on a person, the Arduino and shield run from a small 5 V power bank (≤ 1 A, ≤ 1,000 mAh).
2	**Upload off the person.** Connect the Arduino to a computer only to upload the sketch. Unplug it before recording.
3	**No mains connection of any kind:** no wall charger, no desk supply, no USB cable to a computer while recording.
4	**No diagnosis.** Do not use ardEEG to diagnose, monitor or treat any medical condition.
5	**Implanted devices.** Not on a person with a pacemaker or other implant without a physician's approval.
6	**Skin.** Clean, intact skin only. Stop if it becomes irritated; clean off gel after each session.
7	**Exposed electronics.** No case: handle by the edges, use anti-static precautions, keep liquids away.

Power while recording

Setup	While electrodes are on	Why
Arduino + ardEEG on a small 5 V power bank	SAFE	No path to the mains.
Computer receiving the data over Wi-Fi	SAFE	No cable to the board or the person.
Arduino plugged into a computer's USB	NEVER	Links the person to the mains through the computer.
Arduino on a wall charger	NEVER	Connects the person to the mains through the electrodes.

Good practice

- Charge the power bank **before** the session, never during it.
- Check every cable before each session. An adult or supervisor checks the setup when students use it.
- Clean electrodes with water after use and let them dry; do not soak them in disinfectant or hot water.

02 WHAT YOU NEED

Item	Requirement
ardEEG	Included
Arduino UNO R4 WiFi	The WiFi version — data goes over Wi-Fi
5 V power bank	Small: ≤ 1 A, ≤ 1,000 mAh, charged
USB-C cable	Data cable: sketch upload, then power from the bank
Electrodes	8 dry or wet Ag/AgCl with 2.54 mm connectors + 2 ear clips
EEG cap	10-20 cap
Computer	Python 3 and the Arduino IDE, on the same 2.4 GHz Wi-Fi network
Gel	Only for wet electrodes

03 SETUP IN SEVEN STEPS

1 Find your computer's IP address

Windows: `ipconfig` → IPv4 Address of the Wi-Fi adapter. macOS: System Settings → Wi-Fi → Details. Linux: `ip addr`. Example: 192.168.1.241.

2 Upload the sketch

Install the Arduino IDE and **Arduino UNO R4 Boards**. Open 1.Send_data_via_wifi.ino and set:

ARDUINO SKETCH

```
char ssid[] = "Wifi Name";                 // your 2.4 GHz network
char pass[] = "password";
udp.beginPacket("192.168.1.241", 13900);   // your computer's IP
```

Board **Arduino UNO R4 WiFi** → Upload, then unplug the USB cable.

3 Mount the shield

With the Arduino unpowered, line up all four headers and press down evenly until every pin is seated.

4 Connect the electrodes

Channels 1–8 to the cap, **Ref** to one earlobe, **Biasout** to the other (AE-PIN-004).

5 Power from the battery

Plug the **power bank** into the Arduino. LED D1 lights; the Arduino joins your Wi-Fi and starts sending.

6 Run the Python receiver

TERMINAL

```
pip install matplotlib scipy
git clone https://github.com/Ildaron/ardEEG.git
cd ardEEG/software/python
python 1.Alpha_real_time.py          # set UDP_IP to your computer's IP first
```

A window opens with 8 live graphs.

7 Check the signal

Sit still, relax the face, then close the eyes for 10 seconds: the alpha signal grows, then shrinks when the eyes open.

For research and education only. Not a medical device. Specifications subject to change.

04 FIRST SIGNAL CHECKS

Test	What you should see
Sit still for 30 s	All 8 channels settle to a small, steady signal.
Close the eyes for 10 s	The 8–12 Hz alpha signal grows, then shrinks when the eyes open.
Chew or blink	Large bursts — best seen with a 1–40 Hz filter.

05 TROUBLESHOOTING

Uploading the sketch

Problem	Fix
Board not listed in the IDE	Install Arduino UNO R4 Boards (Boards Manager). Use a USB-C data cable.
`WiFi.h` errors	Choose Arduino UNO R4 WiFi, not the Minima.
Upload fails	Unplug the power bank; use only the computer's USB while uploading. Try another cable.

Wi-Fi and data

Problem	Fix
Python shows nothing	The same IP (your computer's current one) must be in the sketch and in UDP_IP.
Arduino does not join the network	Check the name and password; use a 2.4 GHz network.
Different networks	Both on the same Wi-Fi. Guest networks often block devices from talking to each other.
Firewall blocks the data	Allow Python through the firewall for UDP port 13900.
Address already in use	Another program uses port 13900 (`netstat -ano | findstr :13900` on Windows).
Data slow or with gaps	Run speed_test_250_samples.py; move the Arduino closer to the router.

Signal

Problem	Fix
All channels noisy at once	Check the **REF and BIAS ear clips** first.
One channel flat or very large	That electrode is not touching the skin.
Strong 50/60 Hz hum	The Arduino must run from the power bank, not the computer's USB.
Cannot see blinks or chewing	The default view is 8–12 Hz. Change the filter to 1–40 Hz (AE-SW-006).
Alpha does not rise with eyes closed	Put an electrode on the back of the head (Pz, O1, O2); eyes closed 10 s in a quiet room.

> **NOTE Board or setup?**
> Set all channels to input-shorted (`CHnSET = 0x01`) in the sketch. If the signal stays within a few µV, the hardware is fine.

06 RELATED DOCUMENTS

AE-DS-001 Product Datasheet	**AE-MD-002** Mechanical & Dimensional Drawing	**AE-PCB-003** PCB & Manufacturing Specification
AE-PIN-004 Connector Pinout & Electrode Guide	**AE-SW-006** Data Interface & Software Guide	